生醫量測系統設計
(9920ET5610701)
Design of Biomedical Measurement Systems

Basic concepts of medical instrument

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The content of this ppt is referred to “Medical Instrumentation Application and Design “(Hardcover) by John G. Webster, Fourth Edition

Syllabus

- Generalized medical instrumentation system
- Alternative operational modes
- Medical Measurement Constraints
- Classification of Biomedical Instruments
- Interfering and Modifying Inputs
- Compensation Techniques
- Biostatistics
- Design process for medical instruments

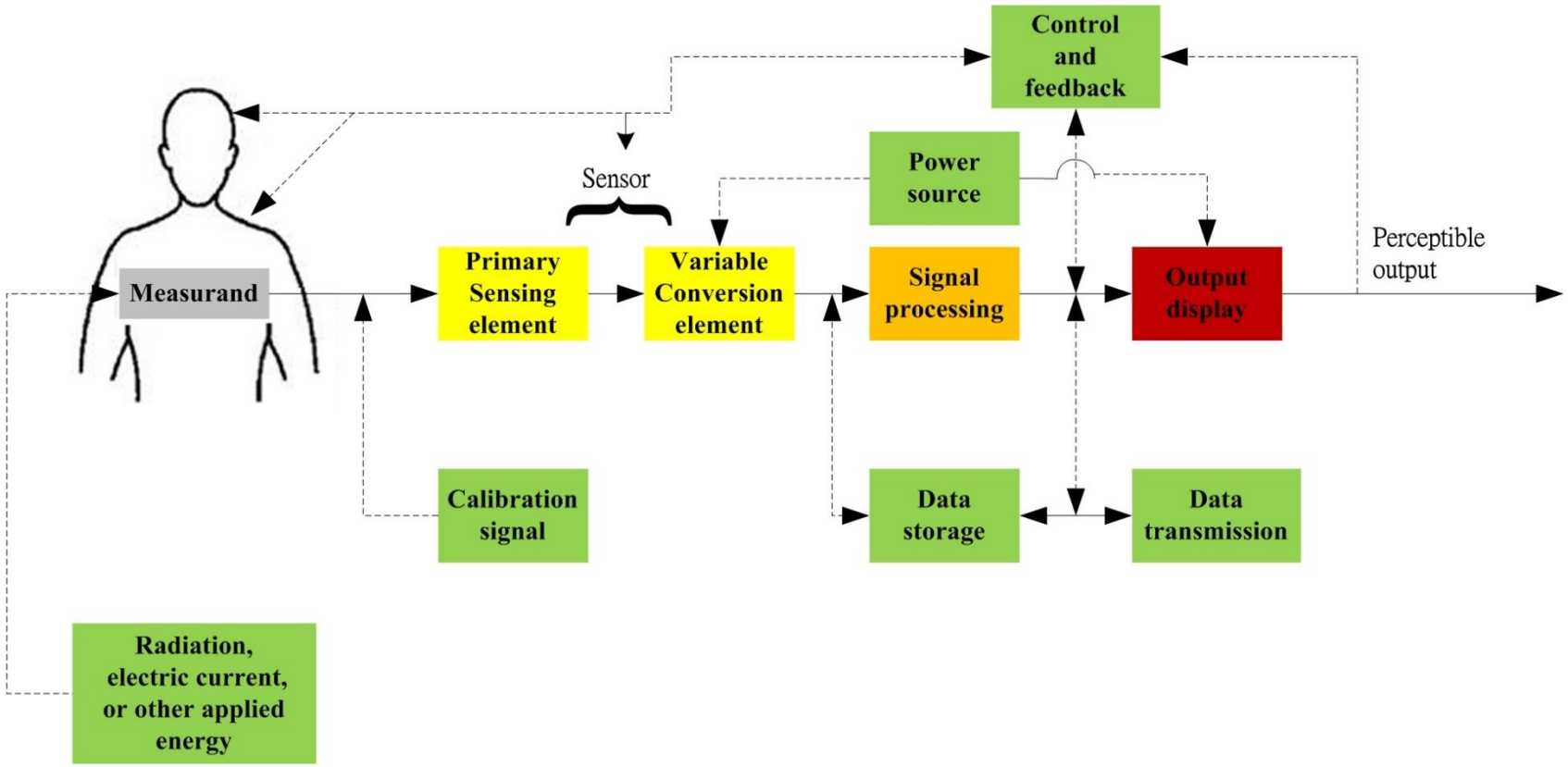
Varied medical devices

TERUMO

3M™ Littmann

FUKUDA FCP-2111 ECG

Generalized medical instrumentation system



Measurand

- The physical quantity, property, or condition that the system measures is called the measurand (被測物理量).
  - Internal (blood pressure)
  - On the body surface (electrocardiogram potential)
  - Emanate from the body (infrared radiation)
  - Removed from the body (blood or biopsy)

Measurand

- Can be grouped in following categories:
  - Biopotential
  - Pressure
  - Flow
  - Dimensions (imaging)
  - Displacement (velocity acceleration, and force)
  - Impedance
  - Temperature
  - Chemical concentrations

Sensor

- Generally, the term transducer is defined as a device that converts one form of energy to another.
- A sensor converts a physical measurand to an electric output.
- The sensor should respond only to the form of energy present in the measurand, to the exclusion of all others.
- Many sensors have a primary sensing element
  - Diaphragm: converts pressure to displacement
- Many variable-conversion elements need external electric power to obtain a sensor output.
  - Strain gage: converts displacement to an electric voltage

Signal conditioning

- Usually the sensor output cannot directly coupled to the display device.
- Simple signal conditioners may only amplify and filter the signal or merely match the impedance of sensor and display.
- Often sensor outputs are converted to digital form and then processed by specialized digital circuits or a microcomputer.
- For example, signal filtering may reduce undesirable sensor signals.

Output display

- The results of the measurement process must be displayed in a form that the human operator can perceive.
- The best form for the display may be numerical or graphical, discrete or continuous, permanent or temporary – depending on the particular measurand and how the operator will use the information. (Doppler ultrasonic signals)
- User controls and the output display should conform to the *Human factors engineering guidelines and preferred practices for the design of medical devices* (AAMI, 1993)

Auxiliary elements

- A calibration signal with the properties of the measurand should be applied to the sensor input or as early in the signal-processing chain as possible.
- Many forms of control and feedback may be required to elicit the measurand, to adjust the sensor and signal conditioner, and to direct the flow of output for display, storage, or transmission.
- Control and feedback may be automatic or manual.
- Data may be stored briefly to meet the requirements of signal conditioning or to enable the operator to examine data that precede alarm conditions.
- Data may be stored before signal conditioning, so that different processing schemes can be utilized.
- Conventional principles of communications can often be used to transmit data to remote displays at nurses' stations, medical centers, or medical data processing facilities.

Alternative operational modes

- Direct-Indirect modes
- Sampling and continuous modes
- Generating and modulating sensors
- Analog and digital modes
- Real-time and delayed-time modes

Direct-Indirect modes

- Often the desired measurand can be interfaced directly to a sensor because the measurand is readily accessible or because acceptable invasive procedures are available.
- When the desired measurand is not accessible, we can use either another measurand that bears a known relation to the desired one or some form of energy or material that interacts with the desired measurand to generate a new measurand that is accessible.
- Examples include
  - cardiac output (volume of blood pumped per minute by the heart), determined from measurements of respiration and blood gas concentration
  - morphology of internal organs, determined from x-ray shadows

Sampling and continuous modes

- Some measurands – such as body temperature and ion concentrations change so slowly that they may be sampled infrequently.
- Other quantities–such as the electrocardiogram and respiratory gas flow may require continuous monitoring.

Generating and modulating sensors

- Generating sensors produce their signal output from energy taken directly from the measurand.
- Modulating sensors use the measurand to alter the flow of energy from an external source in a way that affects the output of the sensor.
- For example,
  - a photovoltaic cell is a generating sensor because it provides an output voltage related to its irradiation, without any additional external energy source.
  - a photoconductive cell is a modulating sensor; to measure its change in resistance with irradiation, we must apply external energy to the sensor.

Analog and digital modes

- Most currently available sensors operate in the analog mode, although some digital measuring devices have been developed.
- Increased use of digital signal processing has required concurrent use of analog-to-digital and digital-to-analog converters to interface computers with analog sensors and analog display devices.
- Quasi-digital sensors, such as quartz-crystal thermometers, give outputs with variable frequency, pulse rate, or pulse duration that are easily converted to digital signal.
- The advantages of the digital mode of operation include
  - greater accuracy, repeatability, reliability, and immunity to noise
  - periodic calibration is usually not required.

Real-time and delayed-time modes

- Sensors must acquire signals in real time as the signals actually occur.
- The output of the measurement may not display the result immediately because some types of signal processing such as averaging and transformations, need considerable input before any results can be produced.
- Often such short delays are acceptable unless urgent feedback and control tasks depend on the output.
- In the case of some measurements, such as cell cultures, several days may be required before an output is obtained.

Medical Measurement Constraints

- The principal measurement and frequency ranges for each parameter are major factors that affect the design of all the instrument components.
  - Low voltages and low frequencies
- Many crucial variables in living systems are inaccessible because the proper measurand–sensor interface cannot be obtained without damaging the system.
- Most measured quantities vary with time, even when all controllable factors are fixed.
- Safe levels of these various types of energy are difficult to establish, because many mechanisms of tissue damage are not well understood.

Medical Measurement Constraints

- Equipment must be reliable, easy to operate, and capable of withstanding physical abuse and exposure to corrosive chemicals.
- Electronic equipment must be designed minimize electric-shock hazards.
- The safety of patients and medical personnel must be considered in all phases of the design and testing of instruments.

Medical and Physiological parameters

Parameter or Measuring Technique	Principal Measurement Range of Parameter	Signal Frequency Range, Hz	Standard Sensor or Method
Electrocardiography (ECG)	0.5-4 mV	0.01-250	Skin electrodes
Electroencephalography (EEG)	5-300 µV	dc-150	Scalp electrodes
(Electrocardiography and brain depth)	10-5000 µV	dc-150	Brain-surface or depth electrodes
Electrogastrography (EGG)	10-1000µV	dc-1	Skin-surface electrodes
	0.5-80 mV	dc-1	Stomach-surface electrodes
Electromyography (EMG)	0.1-5 mV	dc-10,000	Needle electrodes
Eye potentials Electro-oculogram(EOG)	50-3500µV	dc-50	Contact electrodes
Electroretinogram(ERG)	0-900 µV	dc-50	Contact electrodes

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Medical and Physiological parameters

Parameter or Measuring Technique	Principal Measurement Range of Parameter	Signal Frequency Range, Hz	Standard Sensor or Method
Blood flow	**1-300 ml/s**	**dc-20**	**Flowmeter (electromagnetic or ultrasonic)**
Blood pressure, arterial Direct	**10-400 mm Hg**	**dc-50**	**Strain-gage manometer**
Indirect	**25-400 mm Hg**	**dc-60**	**Cuff, auscultation**
Blood pressure, venous	**0-50 mm Hg**	**dc-50**	**Strain-gage**
Blood gases			

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Classification of Biomedical Instruments

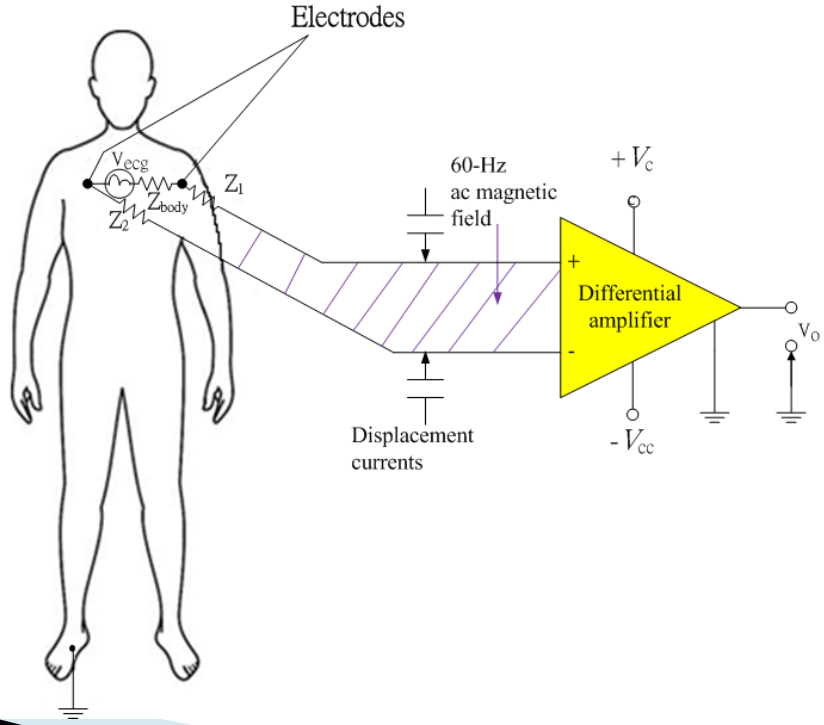
- The study of biomedical instruments can be approached from at least four viewpoints.
  - Techniques of biomedical measurement can be grouped according to the quantity that is sensed, such as pressure, flow, or temperature.
  - A second classification scheme uses the principle of transduction, such as resistive, inductive, capacitive, ultrasonic, or electrochemical.
  - Measurement techniques can be studied separately for each organ system, such as the cardiovascular, pulmonary, nervous, and endocrine systems.
  - Finally, biomedical instruments can be classified according to the clinical medicine specialties, such as pediatrics, obstetrics, cardiology, or radiology.

Interfering and modifying Inputs

- Desired inputs are the measurands that the instrument is designed to isolate.
- Interfering inputs are quantities that inadvertently affect the instrument as a consequence of the principles used to acquire and process the desired inputs.
  - If spatial or temporal isolation of the isolation of the measurand is incomplete, the interfering input can even be the same quality as the desired input.
- Modifying inputs are undesired quantities that indirectly affect the output by altering the performance of the instrument itself.

Interfering and modifying Inputs

- A typical electrocardiographic recording system.
  - The desired input is the electrocardiographic voltage *Vecg that appears* between the two electrodes on the body surface.
  - Two possible interfering inputs are stray magnetic fields and capacitively coupled noise. Orientation of patient cables and changes in electrode-skin impedance are two possible modifying inputs.



Compensation Techniques

- Several compensation methods for eliminating the effects of interfering and modifying inputs.
  - Inherent insensitivity
    - If all instrument components are inherently sensitive only to desired inputs, then interfering and modifying inputs obviously have no effect.
    - Twisting of the electrode wires to reduce the number of magnetic flux lines that cut the shaded loop in last Figure.
  - Negative feedback

Compensation Techniques

- Signal filtering
  - Input filtering blocks interfering and modifying inputs, but does not alter the desired input.
  - Output filtering is usually more difficult because desired and undesired output signals are superimposed.
  - Instruments are often shock-mounted to filter vibrations that affect sensitive instrument components.
  - Electromagnetic shielding is often used to block interfering electric and magnetic fields.
- Opposing inputs

Biostatistics

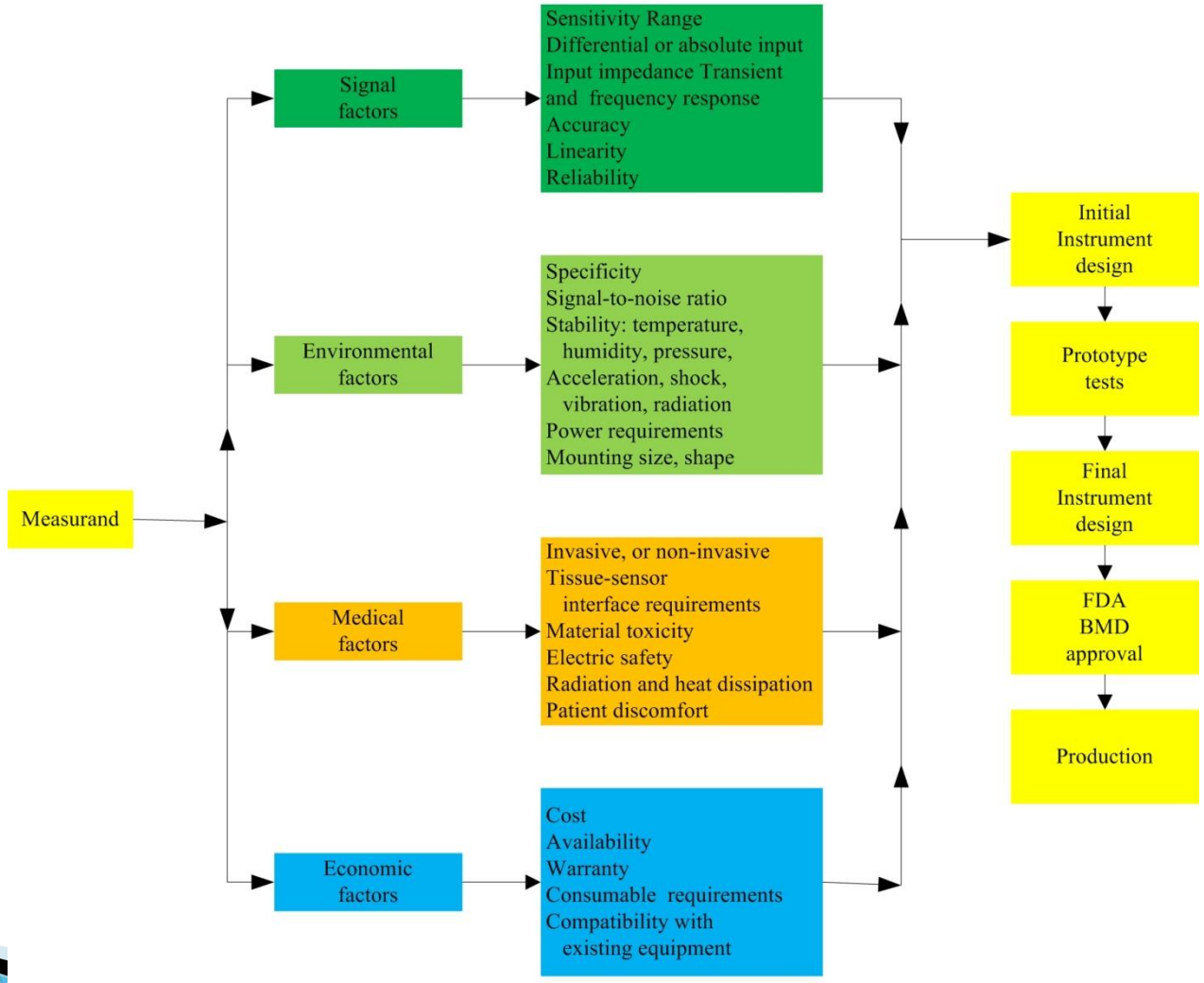
- The application of statistics to medical data is used to design experiments and clinical studies;
  - to summarize, explore, analyze, and present data;
  - to draw inferences from data by estimation or hypothesis testing;
  - to evaluate diagnostic procedures;
  - to assist clinical decision making.
- Mean

$$\bar{X} = \frac{\sum X_i}{n} \qquad (1.4)$$

- Standard deviation

$$s = \sqrt{\frac{\sum (X_i - \bar{X})^2}{n-1}} \qquad (1.6)$$

Biostatistics

- According to the American Association for Medical Instrumentation (AAMI) standards for noninvasive blood pressure monitoring, the bias was no greater than 5 mmHg and the precision no greater than 8 mmHg.
- *Bias* is the average of the differences between the two methods, and the *precision* is the standard deviation of the differences between the two methods.

Design process for medical instruments